

Do-Weighs Forklift application to measure no of components

Current State of the project:

The forklift is equipped with integrated **weigh forks**.

When a user picks up a pallet, the scale displays the **Gross Weight** (Pallet + Components). This hardware integration is now operational.

Going ahead: Do –Weighs application

To determine the exact quantity of components in the pallet, the user will access the **Do-Weighs application**. There can be two scenarios

Scenario I: Unit Weight is known (Already added in master)

Scenario II: Unit Weight is unknown or to be re verified

We need to make arrangements to cater both scenarios in the application

1. **Selection:** User will select the **Item Code** from a searchable List (LOV) or user must be able to scan a barcode/Qr code to enter the item code in the field
2. **Data Fetch:** System to retrieve the **Avg_Unit_Weight** from the Master Database.
3. If user wishes, there must be a provision to override this weight manually during the process.
4. If the Unit Wt. is not available (unknown) in the database master or if the unit Wt. need to be re verified, there must be a provision to calculate the net unit weight.
To calculating the unit Wt. user **will user the second form** "Net Wt. Calculator"
5. User will enter no of components; Gross weight (Kg), pallet Wt. (Kg) and system will calculate the net Unit weight.
In the second from entering pallet Wt. to be kept optional. user can enter "0" or leave blank.
System to calculate unit Wt. accordingly.
6. There should be a provision to save this unit weight to the master report using form 2.
7. **Unit Wt:** User will enter the calculated unit Wt. in form 1
8. **Gross Weight:** User will enter the **Gross Weight** displayed on the forklift scale.
9. **Pallet Wt. -:** User will enter the **Pallet Weight**.
10. **Processing No of components:** System calculates: $\text{Net_Wt} = \text{Gross_Wt} - \text{Pallet_Wt}$,

No of Components = Net_Wt / Unit Weight of the component

11. **"Variance Flag."** If the calculated number of components is not a whole number (e.g., 67.8)
The no of component value to be round off to closest value.
System to show one digit after decimal.

12. **Record Saving:** System displays the **Total no of components** and should asks for user confirmation to save the record.
13. **Clicking on Save button to store the record in the given format**

Data collection and Reports

We will need a common report which will have details of all the components measured on the scale. History with date and time stamp to be maintained.

Sr no	Scanned by Token	Component	Component description	Wt. per component as per item Master (Kg)	Wt. per Component Calculated Manually (Kg)	Gross weight (Kg)	Pallet Weight (Kg)	Net Weight of total components (Kg)	No of Components calculated –Actual	No of Components (Round off)	Date & Time
xx	Fxxxx	XX-XXX-XXXXXX	ABC	4	4.4	800	67	134	166.59	167	05-04-2026 15:30 pm
xx	Fxxxx	XX-XXX-XXXXXX	PQR		2						

• Required dashboards

APPLICATION DASHBOARD 1			
DO-Weights LOGIN PAGE User will Login using their Login ID and Passwords			

FORM 1 – Component Count

APPLICATION FORM 1	
Do-Weights	
Item code	Scan or select item code
Description	MOUNTING FLANGE,25NB (1"),IS2062 GR B
Unit Wt as per system (Mast	4.0
Gross Weight (Kg)	800
Pallet Wt (Kg)	67
No of Components (ACTUAL)	166.59
No of Component Round Of	167
	Save
	Clear

1> LOV selection
2> Entry - via Item barcode scanning

1> To be fetched from system (Master data table)

Details to be Fetched from master table. + Also we need a Provision to overwrite this value manually.

If not a whole number - the value to be rounded off to next closest value .
If value is whole value - show as it is

To save the record in the master report (Report format givrn)

Form 2 – Net wt. Calculator (If Unit Wt. of component is unknown or needs to be verify an existing wt. – the user will use this second form)

APPLICATION FORM 2	
NET Wt. CALCULATOR form	
Item code	
No of components	5
Total Gross Wt (Kg)	89
Pallet Wt (Kg)	67
Unit Weight (Kg)	4.4
Save to report	Clear